# Labels to Generalized Sturmians: a map of the GeoVac papers onto *Hyperspherical Harmonics and Their Physical Applications*

J. Loutey*

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This note maps what the GeoVac papers build on in J. E. Avery and J. S. Avery, *Hyperspherical Harmonics and Their Physical Applications* (World Scientific, 2017), what they add, and where your methods meet a quantum computer. It runs from the discrete object GeoVac starts from, through Fock's projection, up to your generalized-Sturmian secular equation and its molecular form.

Book references give the chapter and section title, which are the stable reference; equation numbers may differ slightly between printings. GeoVac references give the paper and the regression test (under `tests/`) that checks the claim. Every number is quoted from the papers or their tests. Throughout, $K$ is the number of configurations (the dimension of the secular matrix), and the *1-norm* is the sum of coefficient magnitudes when that matrix is written as a weighted sum of unitaries, which sets the cost of qubitization.

**How each connection is labelled.** BUILDS ON; EXTENDS; PUTS NUMBERS ON (the book states it qualitatively and a GeoVac paper measures it); YOURS, CREDITED (results our papers had presented as their own until a direct reading of your book this month, and which now credit you); INDEPENDENT ROUTE; SCOPE OF OUR CLAIM (reading your work narrowed a GeoVac statement).

## 1 The chain, one rung at a time

### 1. The discrete object: a label set with a Casimir on it

Paper 0 starts from an isotropic packing construction. Its shells have capacities $2k-1$, the same counts as the spherical harmonics $Y_{lm}$, and grouping them gives $n^2$ states per shell; the paper records that grouping as an observation, not a consequence of its axioms. The object is the exact label set $(n, l, m)$ of the $S^3$ harmonics, on which the Laplace–Beltrami operator acts as the diagonal SO(4) Casimir,

$$-\Delta_{S^3} = L^2 + A^2, \qquad \text{eigenvalue } n^2-1 \text{ on every label},$$

a classical structure (Pauli 1926, Bargmann 1936) that Paper 7 credits so. It is *not* the lattice's own graph Laplacian: that operator never changes $l$, splits into $n_{\max}$ disconnected blocks, and does not converge to $\Delta_{S^3}$ — Paper 7 proves this and records it as a negative result.

*Independent researcher, `jloutey@gmail.com`. Developed in an AI-assisted workflow (implementation and drafting with Anthropic's Claude); errors are the author's. Papers and tests: https://github.com/jloutey-hash/geovac.

*GeoVac:* Paper 0 abstract; Paper 7 §New contributions; `test_paper7_label_set.py`, `test_graph_s3_nonconvergence.py`.
*Book* (INDEPENDENT ROUTE): Ch. 2, *Degeneracy of hyperspherical harmonics*, $(\lambda+1)^2$ for $d=4$, eq. (2.43); *Eigenfunctions of generalized angular momentum*, eq. (2.8). Paper 0 reaches the count combinatorially.

## 2. Fock's projection: from labels to the hydrogen spectrum

Paper 7 audits Fock's stereographic map and the chordal-distance identity in 18 symbolic checks of the known identities. The $S^3$ Coulomb kernel has eigenvalue $2\pi^2/n$, which forces $p_0 = Z/n$ and gives $E_n = -Z^2/2n^2$ as a function of the Casimir. The labels carry the degeneracies and selection rules; the projection carries the physics. GeoVac reads the two as dual descriptions and takes no position on which is more basic.
*GeoVac:* Paper 7 §Conformal map, §Energy shell; `test_fock_projection.py`, `test_fock_laplacian.py`, `test_paper7_fock_kernel.py`.
*Book* (INDEPENDENT ROUTE): Ch. 5, *Fock's original treatment*, eqs. (5.10)–(5.29). Paper 7 credits Fock 1935 as the primary source for the one-electron chain.

## 3. Coulomb Sturmians at a shared scale

Fix the scale and let the charge vary instead: all functions share one $k$, so all live on the same $S^3$. Papers 8, 34 and 60 use the potential-weighted orthonormality and the one-body matrix in its Sturmian form,

$$\int \chi^*_{n'l'm'}\,\frac{n}{kr}\,\chi_{nlm}\,d^3r = \delta, \qquad h = k^2\big(1-\tfrac{m}{2}\big) - Zk\,\mathrm{diag}(1/n),$$

with $m$ the overlap. GeoVac's production qubit Hamiltonians use hydrogenic orbitals with scale $Z/n$; those are not a Sturmian basis, and Paper 19 says so.
*GeoVac:* Paper 8 §Sturmian basis; Paper 34 §Sturmian projection; Paper 60 eq. (pw); `test_paper60_sturmian.py`.
*Book* (BUILDS ON): Ch. 6, *Potential-weighted orthonormality*, eqs. (6.32), (6.39); Ch. 9, *Matrices representing kinetic energy and nuclear attraction*, eqs. (9.40), (9.44).

## 4. Many electrons

For $N$ electrons the same statement holds on $S^{3N-1}$: the Laplace–Beltrami operator is the diagonal SO($3N$) Casimir $\lambda(\lambda+3N-2)$. Paper 7 cites it as standard hyperspherical-harmonic theory (J. Avery, 1989); Paper 13 uses the SO(6) case for helium, through other references.
*GeoVac:* Paper 7 §N-electron; Paper 13 §Casimir; `test_algebraic_angular.py`.
*Book* (BUILDS ON): Ch. 2, eqs. (2.1), (2.8); the $d=6$ tables in Ch. 3, *Normalization.*

## 5. Generalized Sturmians: the isoenergetic secular equation

Paper 60 takes your secular equation as written. Fix the energy, solve for the scale: the roots are values of $p_\kappa$, each giving an energy directly. There is no overlap matrix and no outer energy search:

$$\big[\,T^0 + T' - p_\kappa \not{V}\,\big]B = 0, \qquad T^0 = Z\,\mathcal{R}_\nu\,\delta, \qquad E = -p_\kappa^2/2.$$

The form for an arbitrary weight potential $V_0$ is also yours, on Goscinski's construction. Earlier versions of Paper 60 presented it as their own; it is now credited:

$$\sum_{\nu} \langle \Phi_\mu | \, V - \beta_\nu V_0 \, | \Phi_\nu \rangle \, C_\nu = 0.$$

*GeoVac:* Paper 60 eq. (secular), eq. (general $V_0$); `test_paper60_sturmian.py`, `test_paper60_general_v0.py`.
*Book* (BUILDS ON; YOURS, CREDITED for eq. (7.13)): Ch. 7, *Sturmians and the many-particle Schrödinger equation*, eq. (7.13); *Derivation of the secular equations*, eqs. (7.78)–(7.89); Goscinski's weighted equation, eq. (7.1).

## 6. The tied scale, measured

In the isoenergetic formulation the basis scale is tied to the eigenvalue. Your text notes that the resulting Goscinskian exponents come out almost exactly at their variational optimum, and that the method suits excited states especially. For the single $1s^2$ configuration the remark is exact: the exponent is the textbook 27/16 and the energy $-2.847$ Ha. Paper 60 measures how the departure grows as configurations are added. For the helium ground state the tied scale stalls at 7.46 mHa while the same span with a free scale reaches 1.28 mHa ($K = 130$). At $K = 452$ the ground state sits 6.82 mHa from exact and the $2\,^1S$ state 1.716 mHa: the cost is state-dependent and falls on the ground state, as your split-shell account explains. Paper 60 also proves a variational upper bound for this formulation, with an inertia corollary, and an interlacing bound; we did not find the upper bound stated in your books or theses.
*GeoVac:* Paper 60 eq. (scale lock), §ground state; `test_paper60_scale_lock.py`, `test_paper60_no_selection.py`.
*Book* (PUTS NUMBERS ON): Ch. 7, closing text of *Derivation of the secular equations*, and the $^1S$ table for helium.

## 7. Two centres: the Shibuya–Wulfman metric returns

For atom-centred Sturmians the metric is weighted by the ket's own nucleus, which makes each centre's block exactly the identity:

$$S_{\tau'\tau} = \frac{n_\tau}{k} \left\langle \chi_{\tau'} \middle| \frac{1}{r_{a(\tau)}} \middle| \chi_\tau \right\rangle = \left\langle \chi_{\tau'} \middle| \frac{p^2 + k^2}{2k^2} \middle| \chi_\tau \right\rangle.$$

Paper 60's regression test reproduces ten of your closed-form Shibuya–Wulfman integrals to about $10^{-12}$. On a full $\sigma$ basis ($n \le N, l < n, m = 0$ on each centre) the conditioning grows exponentially with the number of shells: 18 to $1.1 \times 10^{11}$ from 3 to 15 functions per centre at $kR = 3$. Paper 60 shows why. In this metric each centre's set is complete and the other centre's set is its translate, which is unitary here, so the two-centre metric is a finite section of

$$\begin{pmatrix} I & U \\ U^* & I \end{pmatrix}, \qquad U \text{ unitary}, \quad \text{spectrum } \{0, 2\}.$$

No preconditioner that acts centre by centre can help; that is proved. The remedy you use for the two-centre density expansion, phasing out the smallest eigenvectors, also works on this metric: at threshold $10^{-6}$ it holds the conditioning at $10^6$ with an $H_2^+$ accuracy loss below $10^{-7}$ Ha through 36 functions per centre. The threshold has to fall as the basis grows.

*GeoVac:* Paper 60 §Molecular frontier; `test_paper60_sigma_basis.py`, `test_paper60_sigma_redundancy.py`.
*Book* (PUTS NUMBERS ON): Ch. 8, *Shibuya–Wulfman orbitals*; Ch. 9, *Shibuya–Wulfman integrals*, eq. (9.26), and *Matrices representing kinetic energy and nuclear attraction* (the own-nucleus form), eqs. (9.42)–(9.44); *2-center densities*, the remark after eq. (9.75) that the density-fitting matrix $1 + W$ becomes ill-conditioned through overcompleteness at small $S$, handled by phasing out the smallest eigenvectors. The translation reading itself is Shibuya and Wulfman's (1965).

## 2 On a quantum computer: generated, loaded, paid for

The idea we discussed: the angular structure can be produced on the device from quantum-number labels, so only the radial part has to be loaded, and the radial part depends on one variable. Paper 60 states the idea and tests what it costs. It does not measure the loading cost itself; that remains a motivating argument.

**Generated, from labels.**

- Angular couplings are exact algebraic numbers (square roots of rationals): Gaunt coefficients, $3j$ and $6j$ symbols, computed from $(n, l, m)$ with no quadrature.
- On one centre they are sparse, and the sparsity depends only on $l_{\max}$, not on the potential (Paper 22). Your Ch. 8 says the same of the one-centre coefficients: large but very sparse.
- At two centres $l$ stops being conserved and only the axial $m$ survives (Papers 14, 58). Your Ch. 9 notes that the two-centre density coefficients are not sparse.

**Loaded, one-dimensional.**

- The radial factors depend on one variable: $r$ for an atom, the hyperradius for several electrons, and for two centres closed-form functions of the internuclear distance $R$.
- If the angular part is generated on the device, the loading cost would be a small set of radial seeds rather than an $O(M^4)$ integral tensor.
- For two centres those seeds close over $\{e^a E_1(a),\ \ln,\ \gamma\}$, classically (Sugiura, Roothaan, Ruedenberg); Paper 58 organises them as a closed seed set. At three centres the $(XY|XZ)$ integral over $1s$ orbitals at a common exponent becomes an elliptic Bessel moment, genus one, in your momentum-space setting (Paper 59, which names your 2013 review as the closest prior art); the rungs beyond it, and the proof that they do not depend on the orbitals, are Paper 63.

**Paid for: the 1-norm.**

- What sets the quantum cost is chiefly the 1-norm of the encoded matrix, and for molecules also the metric's conditioning, not the load count alone.
- *Atoms:* the secular matrix carries no metric and needs no energy search. Its 1-norm grows more slowly than its dimension over every basis computed (helium, $K^{0.82}$ over $K = 74$–$164$, local slope falling to 0.77 by $K = 514$), when the basis grows at fixed $l_{\max}$. It is bought with the same tied scale that sets the accuracy floor: 6.82 mHa for the ground state at $K = 452$, against 1.716 mHa for $2\,^1S$ at the same encoding cost.

- *Molecules:* the metric returns and the 1-norm is ordinary polynomial. The metric is paid once, when the one-electron molecular orbitals are built; it does not compound with electron number.

In short: your isoenergetic formulation makes the atomic problem a clean quantum target at a state-dependent accuracy floor, a trade-off between cost, accuracy and which state is targeted rather than an unconditional efficiency gain. A targeted search did not find this combination in the existing quantum-algorithm literature. For molecules, the cost moves into the Shibuya–Wulfman metric, and rung 7 above is where that cost now stands.

# 3 Concordance: your book and the GeoVac papers

| Your book | GeoVac | Relation | Checked by |
|---|---|---|---|
| Ch. 2, generalized angular momentum, eq. (2.8) | Paper 7, SO($3N$) Casimir (§N-electron; cites J. Avery 1989) | builds on | `test_algebraic_angular` |
| Ch. 2, degeneracy, $(\lambda+1)^2$, eq. (2.43) | Paper 0, $n^2$ shell grouping, reached combinatorially | independent | `test_paper7_label_set` |
| Ch. 5, Fock's original treatment, eqs. (5.10)–(5.18) | Paper 7, stereographic map and chordal identity (credited to Fock 1935) | independent | `test_fock_projection` |
| Ch. 5, kernel eigenvalue and $Z/(kn)=1$, eqs. (5.21)–(5.29) | Paper 7, kernel eigenvalue $2\pi^2/n$ forces $p_0 = Z/n$ | independent | `test_paper7_fock_kernel` |
| Ch. 5, Coulomb Sturmians, eqs. (5.32), (5.33) | Paper 34, Sturmian reading of the labels at $\lambda = Z/n$ | builds on | `test_paper34_projection_spot_checks_batch1` |
| Ch. 8 and Ch. 9, the four-dimensional three-harmonic coefficient | Papers 32, 38, 44, 45, the Wen–Avery 3-$Y$ integral on $S^3$ | builds on | `test_so4_three_y_integral` |
| Ch. 6, potential-weighted orthonormality, eqs. (6.32), (6.39) | Paper 60 eq. (pw); Paper 8 Sturmian orthogonality | builds on | `test_paper60_sturmian` |
| Ch. 9, kinetic-energy and nuclear-attraction matrices, eqs. (9.40), (9.44) | Paper 34, $h = k^2(1-m/2) - Zk\,\mathrm{diag}(1/n)$; Paper 60, kinetic matrix $\nVdash -$ (overlap)/2 | builds on | `test_paper60_scale_lock` (Paper 60 form) |
| Ch. 7, Goscinski's weighted equation and orthogonality, eqs. (7.1), (7.5)–(7.8) | Paper 60 §Isoenergetic reformulation | builds on | `test_paper60_general_v0` |
| Ch. 7, many-particle Schrödinger equation, eq. (7.13) | Paper 60, secular equation for a general weight $V_0$ (now credited to you and Goscinski) | yours, credited | `test_paper60_general_v0` |

| Your book | GeoVac | Relation | Checked by |
|---|---|---|---|
| Ch. 7, derivation of the secular equations, eqs. (7.78)–(7.89) | Paper 60 eq. (secular) | builds on | `test_paper60_sturmian` |
| Ch. 7, Goscinskian configurations, eqs. (7.64)–(7.77) | Paper 60: He $1s^2 = -2.847$ Ha reproduced; a domain rule for large $n$ | builds on | `test_paper60_box_rule` |
| Ch. 7, near-optimal exponents; suited to excited states | Paper 60: ground state 7.46 mHa tied vs 1.28 mHa free ($K = 130$); at $K = 452$, 6.82 vs 1.716 mHa for $2\,^1S$ | puts numbers on | `test_paper60_scale_lock` |
| Ch. 7, tied-scale formulation | Paper 60: variational upper bound (inertia corollary) and interlacing bound; upper bound not found in your books or theses | extends | `test_paper60_scale_lock`, `test_paper60_no_selection` |
| Ch. 7, molecular spectra, eqs. (7.48)–(7.63) | Paper 60, identity-form molecular secular equation; $T^0$ diagonal for molecular Sturmians (now credited to you) | yours, credited | — |
| Ch. 8, Shibuya–Wulfman orbitals: fixed-$S$ standard eigenproblem | Paper 60: fixed-$S$ vs fixed-$R$ on $H_2^+$, $8.5 \times 10^{-2}$ vs $8.5 \times 10^{-4}$ Ha at 3 functions per centre, same basis | puts numbers on | `test_paper60_sigma_basis` |
| Ch. 9, Shibuya–Wulfman integrals, eqs. (9.26), (9.27) | Paper 60 eq. (sw), Hermitian form; ten closed forms reproduced to $\sim 10^{-12}$ | builds on | `test_paper60_sigma_basis` |
| Ch. 9, nuclear-attraction form, own-nucleus weight, eqs. (9.42), (9.43) | Paper 60: the metric is weighted by the ket's own nucleus (corrects our earlier "the metric is $V_0$") | yours, credited | `test_paper60_sigma_redundancy` |
| Ch. 8, Shibuya–Wulfman translation construction | Paper 60: two-centre metric a finite section of $\left(\begin{smallmatrix} I & U \\ U^* & I \end{smallmatrix}\right)$, spectrum $\{0, 2\}$; no centre-wise preconditioner helps (translation reading credited to Shibuya–Wulfman 1965) | extends | `test_paper60_sigma_redundancy` |
| Ch. 9, 2-center densities: $1 + W$ ill-conditioned by overcompleteness; phase out the smallest eigenvectors (text after eq. (9.75)) | Paper 60: the same remedy applied to the Shibuya–Wulfman metric on the $\sigma$ basis; at $10^{-6}$ it keeps $H_2^+$ within $10^{-7}$ Ha through 36 functions per centre | puts numbers on | `test_paper60_sigma_redundancy` |
| Ch. 8 and Ch. 9, one-centre coefficients large but very sparse | Paper 14 Gaunt selection rules; Paper 22 sparsity depends only on $l_{\max}$ | independent | Paper 22 / 14 suites |
| Ch. 9, two-centre density coefficients not sparse | Papers 14 and 58: $l$ selection lost at two centres, axial $m$ survives | independent | Paper 14 suite |
| Ch. 9, interelectron repulsion between one-centre densities, momentum space, eqs. (9.52)–(9.60) | Paper 59: the $(XY\|XZ)$ three-centre integral over $1s$ at a common exponent is an elliptic Bessel moment, genus one (names your 2013 review as closest prior art) | extends | `test_paper59_*` |

| Your book | GeoVac | Relation | Checked by |
|---|---|---|---|
| App. B, Coulomb-Sturmian expansion of functions of $s = kr$, eq. (B.5); STO integrals | Paper 58: these integrals are standard Coulomb–Sturmian practice, which GeoVac does not add to | builds on | — |
| Ch. 8 and Ch. 9, cross-centre overlap couples $n$ | Paper 8: our no-binding theorem holds only for a single-$n$ rotation approximation; the genuine Sturmian overlap escapes it and binds | scope of our claim | `test_paper8_overlap_cross_n` |
| Ch. 7, generalized Sturmians, eq. (7.1) | Paper 8 structural theorem, scoped to the approximation, not to your basis | scope of our claim | `test_paper8_sturmian_structural` |
| Ch. 2, hyperradial separation, eq. (2.1) | Paper 16, kinetic operator in hyperspherical form | builds on | — |
| Ch. 3, *Normalization*, $d = 6$ tables | Paper 13, SO(6) Casimir for helium | independent | `test_algebraic_angular` |
| Ch. 9, Shibuya–Wulfman integrals, own-nucleus form, eq. (9.43) | Paper 19 §Shibuya–Wulfman integrals | builds on | — |

# 4 Where it is open

**The molecular secular equation as a quantum target.**
Your *Molecular spectra* (Ch. 7) gives the identity-form secular equation for configurations built from molecular generalized Sturmians, with the weights $\beta_\nu$ found by interpolation. Paper 60 leaves the molecular 1-norm open with the obstruction identified: in the configuration-scaled form the nuclear-attraction matrix is neither diagonal nor independent of $p_\kappa$ on two centres, and the one measured molecular 1-norm, from an ordinary two-centre CI, grows polynomially. Whether your molecular-Sturmian form avoids that obstruction is not tested.

**A schedule for phasing out eigenvectors.**
Phasing out works on the $\sigma$ basis. Each fixed threshold tested between $10^{-2}$ and $10^{-4}$ stalls once the basis grows, while $10^{-6}$ holds through 36 functions per centre. Paper 60 concludes the threshold must fall with the basis; a shrinking schedule has not been tested.

**Lowering the ground-state floor.**
Paper 60 takes the mechanism from you: in $1s^2$ both electrons share one exponent, so the in-out correlation that a split-shell $1s\,1s'$ pair carries is missing. What it leaves open is whether a weighting potential of a different radial shape could lower the ground-state floor while keeping the Fock mapping.

**Three centres.**
The three-centre two-electron integral $(XY|XZ)$, over $1s$ orbitals at a common exponent, is genus one: an elliptic Bessel moment in the momentum-space Slater theory of your Ch. 9 (Paper 59, which names your 2013 review as the closest prior art). Paper 59 identifies the class; a closed form is not known.

**The rungs beyond, and a question for you.**

Paper 63 follows the class upward. Any tree of $n$ linked two-centre densities has one holomorphic period, a Bessel moment $\int \prod_v K_0(a_v x)\, dx$ in $n$ scales, whose one-scale operator is the Legendre operator Hadamard the equal-mass banana Feynman integral with $n-1$ propagators: elliptic at two densities, a K3 surface at three, a Calabi–Yau threefold (entry #58 of the published tables) at four, nothing new at five. Every exponential-type orbital is a derivative of the $1s$ Yukawa function in its exponent and its centre, so the ladder does not depend on the orbitals, only on the density graph. A Coulomb ring of three does not climb; a geminal ring does; a Gaussian geminal leaves the ladder. Your Ch. 9 and App. B reach multicentre repulsion integrals by expanding each two-centre density in a $2k$ Coulomb–Sturmian basis on its two centres, a series whose short-range ill-conditioning your 2013 paper handles by phasing out small eigenvalues; in Paper 63's reading that series meets the genus jump at short range, since an elliptic period is no finite combination of genus-zero terms. The question we would most like to put to you: does your generalized Shibuya–Wulfman / hyperspherical-harmonic expansion of mixed-scale repulsion contain the Bessel-moment reduction in another form? Our reading of the monograph says no; you would know.

## Papers referred to

All under `papers/` at https://github.com/jloutey-hash/geovac.

- Paper 0, *Counting Angular Momentum Labels by Isotropic Packing*
- Paper 7, *The Dimensionless Vacuum*
- Papers 8–9, *Bond Sphere*
- Paper 13, *The Hyperspherical Lattice*
- Paper 14, *Structurally Sparse Qubit Hamiltonians*
- Paper 16, *The Periodic Table Recast as $S_N$ Representation Theory*
- Paper 19, *Toward PK-Free Molecular Hamiltonians*
- Paper 22, *Potential-Independent Angular Sparsity*
- Papers 32, 38, 44, 45, the spectral-triple papers that use the Wen–Avery integral
- Paper 34, the projection catalogue
- Paper 58, *Angular Sparsity Is an Atomic-Sector Property*
- Paper 59, *The Three-Center ERI Is an Elliptic Bessel Moment*
- Paper 60, *The Generalized-Sturmian Secular Equation Block-Encodes Without a Metric*
- Paper 63, *Linked Electron Densities Climb the Banana Ladder*